

Jane Street Strategy & Product Superday

Market-Structure, ETF, LULD, Risk, Settlement, and Regulation Study Guide

Expanded two-day technical prep version

Prepared for Alex Zhang

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How to use this guide in the last two days

Anthony's main warning was that the reading packet is not the whole syllabus. Treat the packet as the intro page. The interview can move from ETFs into exchange mechanics, auctions, LULD, market-maker exposure, reopening failures, settlement, abnormal data, or regulatory design. In the transcript, Anthony said the key is knowing how to build an exchange, how Jane Street sits in that exchange, how the flows work, and only then the ETF product itself. He also warned that generic consulting frameworks do not solve these questions; either the underlying market mechanics are understood, or the answer becomes vague.

Two-day priority order:

1. Master exchange mechanics: order book, matching, auctions, market data, order lifecycle.
2. Master LULD mechanics and auction/reopening edge cases.
3. Master ETF fair value, AP creation/redemption, market-maker hedging, and why arbitrage fails.
4. Master August 24, 2015 and RSP as an integrated market-structure failure.
5. Master Anthony-style scenarios: levered ETP, client block, lower-band pause, what you know/do not know, desk exposure.
6. Master trade settlement risk, abnormal data invariants, and the SEC rule cheat sheet.

Interview answer. When stuck, start from layers: exchange mechanics → order book/auction → market maker/AP behavior → ETF fair value → risk/edge case. Do not start from memorized facts.

1 Master map of the interview universe

Everything reduces to seven layers:

Layer	What you must understand
Exchange layer	Orders, books, matching, auctions, halts, LULD, market data, execution reports.
National market system	Venues, SIP, NBBO, protected quotes, routing, trade-through prevention, market centers.
Participant layer	Retail/institutional flow, brokers, market makers, APs, lead market makers, clearing members.
ETF layer	ETF market price, NAV, iNAV/IIV, fair value, AP creation/redemption, premium/discount.
Market-maker layer	Quoting, hedging, inventory, adverse selection, block facilitation, basis risk.
Risk layer	Market, liquidity, credit, settlement, operational, model, basis, and regulatory risk.
Regulatory layer	LULD, Reg NMS Rule 611, Reg SHO, Rule 15c3-5, Rule 605/606, historical Rule 48.

1.1 One paragraph master answer

An exchange collects buy and sell orders, maintains a limit order book, matches crossing orders by deterministic rules, and publishes market data and execution reports. An ETF trades on that exchange like a stock, but its economic value is linked to an underlying basket. Market makers and authorized participants keep ETF price near fair value by pricing, hedging, and arbitraging the ETF against that basket or related hedges. During stress, the arbitrage loop can break if the basket is not open, is halted, is stale, has wide spreads, or if hedges such as futures are unavailable. LULD protects individual securities from trading outside dynamic bands, but it does not guarantee cross-instrument consistency between ETF, components, futures, and similar products.

Trap. Do not say “the ETF price was fake.” The ETF market price was real executable market price. It was just a poor measure of underlying fair value during the dislocation.

2 Exchange mechanics from first principles

2.1 What is an exchange?

An exchange is a deterministic matching system. Participants send orders. The exchange validates and sequences them. The matching engine applies rules to determine whether they trade. The order book stores unfilled orders. The exchange publishes public market data and private execution reports.

Participants / brokers / market makers

- > order entry gateway
- > validation
- > sequencer
- > matching engine
- > order book + trade event log
- > market data + private execution reports

The exchange does not decide fundamental value. It decides which orders trade with which other orders under published rules.

2.2 Core order fields

A minimal exchange order needs only the fields required to identify, prioritize, and execute it:

Field	Why it matters
Order ID	Uniquely identifies the order for fills, cancels, and deduplication.
Participant/session ID	Who submitted it; needed for permissions and reports.
Symbol	What instrument is being traded.
Side	Buy or sell.
Quantity	How much.
Price	Limit price; absent for pure market order.
Order type	Market, limit, stop, etc.
Time-in-force	Day, IOC, FOK, GTC, auction-only, etc.
Timestamp/sequence	Needed for priority and event reconstruction.

Trap. Do not confuse side and order type. “Buy” is side. “Limit” is order type.

2.3 Order book, bid, offer, spread, and depth

A bid is an order to buy. An offer or ask is an order to sell. The order book is the set of resting unfilled buy and sell orders.

XYZ book

Bids		Offers	
Qty	Price	Price	Qty
100	49.90	50.10	100
200	49.80	50.20	300
300	49.70	50.30	200

$$\text{Spread} = \text{Best Offer} - \text{Best Bid}$$

$$\text{Mid} = \frac{\text{Best Bid} + \text{Best Offer}}{2}$$

Depth is quantity available at each price level. Thin depth means little quantity is available near the current price, so aggressive orders can move price quickly.

Interview answer. A large market sell order does not execute at theoretical fair value. It executes against the bids actually resting in the book. If the book is thin, the order walks down through worse and worse prices.

2.4 When does a trade happen?

For a buy limit order, a trade can happen when:

$$\text{Buy limit price} \geq \text{Best offer.}$$

For a sell limit order, a trade can happen when:

$$\text{Sell limit price} \leq \text{Best bid.}$$

This is called crossing the spread.

2.5 Price-time priority

Most simple equity books use price-time priority:

1. Better price wins.
2. If price is equal, earlier exchange sequence wins.

For bids, higher price is better. For offers, lower price is better.

Edge case. If two same-price orders arrive nearly together, the matching engine does not debate human timestamps. The exchange sequencer assigns an official order, and that sequence determines priority.

2.6 Market orders, limit orders, stop orders, and stop-limit orders

Type	Meaning	Main danger
Market	Trade immediately at available prices.	Guarantees urgency, not price; can walk a thin book.
Limit	Trade only at limit price or better.	May not fill.
Stop-loss	Becomes market order after trigger price.	Trigger does not guarantee execution price; dangerous in gaps.
Stop-limit	Becomes limit order after trigger.	Avoids terrible price but may not fill.
Auction-only	Participates only in opening/closing/reopening auction.	Execution depends on auction clearing.

Interview answer. Market orders are dangerous in stressed ETF markets because they demand immediate liquidity exactly when market makers may have widened or pulled quotes. A stop order can become a market order at the worst time.

2.7 Time in force

TIF	Meaning
Day	Rests until filled or end of day.
IOC	Immediate or cancel: fill what is available now; cancel the rest.
FOK	Fill or kill: fill entire size immediately or cancel all.
GTC	Good till cancelled: remains until filled/cancelled under venue rules.
MOO/LOO	Market-on-open / limit-on-open: auction orders for open.
MOC/LOC	Market-on-close / limit-on-close: auction orders for close.

2.8 Partial fills and remaining quantity

An order can fill partially. Track original quantity, cumulative filled quantity, remaining quantity, and status:

$$\text{Remaining} = \text{Original} - \text{Cumulative Fills} - \text{Cancelled Quantity.}$$

If a buy order for 400 at 50.20 finds only 300 shares offered at 50.20 or lower, then 300 fills and 100 either rests or cancels depending on the order's time-in-force.

2.9 Cancel and replace

A cancel request is not the same as a cancellation. It is a request processed in exchange sequence.

Live order

- > cancel request sent
- > cancel pending
- > cancel confirmed, cancel rejected, or fill arrives first

Edge case. If a fill and cancel cross, whichever reaches the matching engine first wins. Until cancel confirmation, the order can still trade.

A replace or modify is not magic. In a safe mental model:

replace = cancel old order + account for fills + send new order

Only treat it as atomic if the exchange explicitly supports an atomic cancel/replace and confirms the new state.

2.10 Public market data and private reports

Public market data:

- Top of book: best bid, best offer, sizes.
- Depth: quantities at multiple price levels.
- Trades: last sale, size, timestamp.
- Status: halted, auction, limit state, open/closed.

Private execution reports:

- Accepted/acknowledged.
- Rejected.
- Partial fill.
- Full fill.
- Cancel confirm.
- Cancel reject.
- Expired.

2.11 SIP, direct feeds, NBBO, protected quotes

The Securities Information Processor (SIP) consolidates market data across venues. Direct feeds come directly from exchanges and can be faster. The NBBO is the national best bid and offer across protected venues.

Term	Meaning
NBB	National best bid: highest displayed protected bid.
NBO	National best offer: lowest displayed protected offer.
NBBO	Combined NBB/NBO.
Protected quote	Automated displayed quote meeting Reg NMS protection requirements.
Trade-through	Execution at worse price than a protected quote elsewhere, absent exception.
ISO	Intermarket sweep order; used when routing sweeps protected quotes elsewhere.

Trap. NBBO is not the same as fair value. It is a displayed quote construct. Fair value is an economic estimate. Last sale is a transaction. Reference price is a rule input.

2.12 What Jane Street does in this system

Jane Street can be a market maker, liquidity provider, AP, block facilitator, and proprietary trading firm. In the exchange layer it is a participant, not the exchange. It sends new orders, cancels, and replaces. It receives market data and fills. Its job is to price, quote, hedge, manage inventory, and avoid adverse selection.

Interview answer. The exchange processes orders by rule. Jane Street decides what orders to send. The market maker's edge comes from pricing and risk management, not from the exchange giving it special execution rules.

3 Auctions: open, close, and reopen

3.1 Why auctions exist

Continuous trading works well when buyers and sellers arrive steadily. But at the open, close, and after halts, there can be a backlog of orders and uncertainty. A call auction batches interest and finds one clearing price.

3.2 Auction clearing price

A simplified auction chooses the price that maximizes executable volume, then uses tie-breakers such as minimizing imbalance and proximity to reference price.

Buy interest	Sell interest
Buy 1000 @ 100.00	Sell 500 @ market
Buy 800 @ 99.50	Sell 700 @ 99.50
Buy 500 @ 99.00	Sell 1200 @ 100.00

At each candidate price, compute executable buy quantity and executable sell quantity. Executable volume is the smaller of the two. The auction price is the price that clears the most volume subject to venue rules.

3.3 Indicative price

Indicative price is the exchange's current estimate of where the auction would clear given current auction interest. It is useful information, not a guaranteed executable price.

Interview answer. Indicative price is not a firm trade price. It is a theoretical auction clearing estimate based on current supply/demand. If a client wants to trade during a halt at the indicative price, we can choose to warehouse risk at a negotiated price, but we should not describe the indicative price as executable market price.

3.4 Auction imbalance

Imbalance means buy and sell interest do not match at the indicative price. Large sell imbalance means more sellers than buyers at the current indicative price. In a stressed reopening, imbalance and indicative price can change rapidly as orders enter/cancel.

3.5 Opening auction

At the open, the primary listing exchange runs an opening process to establish an official opening price. A stock does not automatically enter continuous trading at 9:30:00. If there is a large imbalance, uncertain clearing price, or operational issue, the open can be delayed.

Interview answer. Many stocks were not open at the start of August 24 because their opening auctions had not established clean prices. The exchange was trying to clear large imbalances under extreme volatility, so ETF market makers lacked reliable live prices for many constituents.

3.6 Closing auction

The close matters because many funds, ETFs, benchmarks, and risk systems use official closing prices. Closing auctions concentrate volume and can dominate end-of-day liquidity.

3.7 Reopening auction after halt

After a trading pause, the primary exchange generally tries to reopen through an auction. Reopening is hard because the market must discover a new price while participants are uncertain about latent demand, stale orders, hedges, and fair value.

3.8 Why reopening can fail

An auction can fail if the indicative clearing price is outside the reopening collar, if imbalance is too large, if there is insufficient eligible interest, or if the primary exchange has a systems issue.

Edge case. If the ETF is paused but its underlier keeps trading, the indicative reopening price may chase the underlier. But it still is not an executable price until the auction prints.

3.9 Auction collars

Auction collars limit how far an auction price can be from a reference such as last sale or reference price. Anthony mentioned collars around 3%, 5%, and 10% from last sale in some venue contexts. The exact venue rule can vary; the interview goal is to understand why a collar exists and why it can cause reopening failure.

Interview answer. A reopening collar prevents an auction from clearing at an extremely far price without another pause. The tradeoff is that if the real market-clearing price is outside the collar, the auction fails and the pause extends.

4 LULD: Limit Up-Limit Down

4.1 Purpose

LULD is designed to prevent trades in NMS securities from occurring outside dynamic price bands around a reference price. It is a single-security volatility control, not an ETF-NAV control. Official LULD materials state that if a limit state is not exited within 15 seconds, the primary listing exchange declares a trading pause, generally five minutes.¹

4.2 Reference price

The basic mechanics:

$$\text{Upper Band} = \text{Reference Price}(1 + p)$$

$$\text{Lower Band} = \text{Reference Price}(1 - p)$$

where p is the percentage parameter.

In the Jane Street packet framework, the reference price is based on recent eligible trades, usually a five-minute arithmetic mean, with update rules. If no regular opening trade exists, the previous close can matter depending on the rule path.

Trap. Reference price is not necessarily last sale, fair value, NAV, or NBBO midpoint. Keep those concepts separate.

4.3 Tier and price buckets

Simplified packet table:

Security/price bucket	Tier 1	Tier 2	Intuition
> 3	5%	10%	More liquid names get tighter bands.
0.75 to 3	20%	20%	Low price means more percentage noise.

¹Official LULD Plan website: <https://www.luldplan.com/>.

< 0.75	lesser of 0.15 or 75%	same	Penny-like names need special treatment.
Leveraged ETP	multiply by leverage factor	multiply by leverage factor	Levered products naturally move more.

Tier 1 includes S&P 500, Russell 1000, and selected ETPs in the packet. Tier 2 is generally other NMS stocks/ETPs.

4.4 Time-of-day multiplier

In the packet logic, percentage parameters are doubled in the first 15 minutes and last 25 minutes of regular trading. SEC research on August 24 notes that LULD bands are doubled during the first 15 minutes of regular trading hours.²

4.5 Leveraged ETP band examples

Example 1: Tier 2 3x ETP, reference price 30, normal time.

$$p = 10\% \times 3 = 30\%$$

$$\text{Band} = 30(1 \pm 0.30) = [21, 39]$$

Example 2: Same product at 9:40, if doubled window applies.

$$p = 10\% \times 3 \times 2 = 60\%$$

$$\text{Band} = 30(1 \pm 0.60) = [12, 48]$$

Interview warning: if a product is Korean-listed, local rules may differ. If the interviewer frames it as a U.S. LULD/NMS hypothetical, apply the U.S. LULD logic.

4.6 Limit state

A security enters a Limit State if the national best offer equals the lower band or the national best bid equals the upper band. If it remains in Limit State for 15 seconds, a trading pause is triggered.

Normal trading

- > NBO equals lower band or NBB equals upper band
- > Limit State for up to 15 seconds
- > if not cured, primary declares trading pause
- > reopening auction

4.7 Straddle state

Straddle State is an edge condition where the national best quote is outside a LULD band but the market is not in Limit State. For example, NBB below lower band or NBO above upper band. Primary exchange can declare a pause in abnormal conditions.

Edge case. You do not need to lead with Straddle State, but knowing it helps if the interviewer asks what happens when quotes are outside bands but not exactly at the band.

4.8 What LULD can do and cannot do

Can do	Cannot do
Stop trades outside individual-security bands.	Determine fundamental value.
Pause trading if a security is stuck at a band.	Force ETF price to equal NAV/iNAV.

²SEC Research Note, *Equity Market Volatility on August 24, 2015*: https://www.sec.gov/marketstructure/research/equity_market_volatility.pdf.

Give liquidity providers time to update.	Ensure ETF constituents are open and tradable.
Coordinate reopening through primary exchange.	Guarantee hedge availability.
Reduce unlimited book-walking.	Eliminate bad executions before pause is triggered.

Trap. Do not say LULD caused August 24. Say it interacted with stress by creating asynchronous halts and reopenings across ETFs, constituents, and hedges.

4.9 Open, reopen, close, and exemptions

Opening: before the primary open, special rules apply. If a security has not opened, bands may not yet apply in the normal continuous way. The opening auction becomes important because it establishes the first official reference point.

Reopening: after a pause, the primary attempts reopening auction. Indicative price and imbalance are informative but not executable.

10-minute failure path in the packet: if the primary cannot reopen due to system issue or no reopening occurs within 10 minutes after the pause begins, other trading centers may resume trading; prior reference price becomes reference; bands are widened to three times normal for 30 seconds, then revert. This is packet-specific and exactly the kind of edge Anthony pushed.

Late-day pause: if a LULD pause is declared near the close, the primary may move toward closing procedures rather than trying to reopen continuous trading.

4.10 LULD vs other market protections

Mechanism	Different purpose
LULD	Single-security dynamic bands and pauses.
Market-wide circuit breaker	Broad index decline pauses entire market.
Futures limit	Futures-specific limit mechanics; can affect hedge availability.
Auction collar	Limits auction clearing price relative to reference/last sale.
Reg NMS Rule 611	Prevents trade-throughs of protected quotations, not volatility.
Market Access Rule	Broker-dealer pre-trade risk controls, not exchange price bands.

5 ETF mechanics

5.1 ETF market price vs NAV vs iNAV vs fair value

Concept	Meaning
Market price	Actual exchange price where ETF shares trade.
NAV	Official net asset value, typically end-of-day based on holdings valuation.
iNAV/iIV	Intraday indicative value, an estimate based on current component data.
Fair value	Market maker's live economic estimate, possibly using basket, futures, ADRs, FX, similar ETFs, and models.
Premium/discount	$(P_{ETF} - FV)/FV$ or $(P_{ETF} - NAV)/NAV$ depending on reference.

Trap. An ETF can have a real market price that is far from fair value. The trade is real; the price signal is distorted.

5.2 ETF fair value

For a simple equity ETF:

$$FV_t \approx \sum_i w_i P_{i,t} + \text{cash} + \text{accruals} - \text{fees}.$$

For international ETFs, add FX and time-zone issues. For derivative-based ETFs, use futures/swaps/options marks and collateral.

5.3 Authorized participants and creation/redemption

Creation:

AP delivers basket/cash to ETF issuer
 -> receives creation unit of ETF shares
 -> can sell ETF shares in secondary market

Redemption:

AP delivers ETF shares to issuer
 -> receives underlying basket/cash
 -> can sell/hedge underlying securities

This keeps ETF price near fair value in normal markets. If ETF is rich, create and sell ETF. If ETF is cheap, buy ETF and redeem for basket.

5.4 Why AP arbitrage is not instant free money

Creation/redemption is not a magic button. It has:

- execution risk in ETF and basket,
- timing/settlement risk,
- funding and borrow costs,
- basket composition and corporate action issues,
- minimum creation unit size,
- operational cutoffs,
- hedge basis risk,
- stale or uncertain fair value.

Expected P&L = $FV - P_{ETF}$ – hedge cost – execution risk – funding – model uncertainty – liquidity risk.

Interview answer. ETF arbitrage is not a law of physics. It is a trade requiring reliable fair value, executable hedges, balance sheet, and operational capacity.

5.5 ETF order book vs stock order book

Dimension	Single stock	ETF
Value anchor	Company-specific value and supply/demand.	Basket fair value and arbitrage capacity.
Liquidity source	Direct order book in the stock.	ETF book plus underlying basket/hedges.
Natural flow	Company investors, funds, retail.	Market makers/APs can dominate, especially niche ETFs.
Stress failure	Stock gaps, halts, spreads widen.	ETF decouples if basket/hedge unreliable.
Displayed volume	Often indicative of direct liquidity.	ETF may have low volume but deep underlying liquidity in calm markets.

Interview answer. A stock book is about supply and demand for one security. An ETF book is partly a synthetic market built from the underlying basket. If the basket cannot be priced or hedged, ETF displayed liquidity can vanish.

5.6 Market maker's ETF loop

Receive market data for ETF + components + futures

- > compute fair value and confidence
- > quote ETF bid/offer around fair value
- > if filled, hedge with basket/futures/similar ETF
- > manage inventory and basis risk
- > adjust quotes if market state changes

Market makers widen spreads when:

- fair value uncertainty rises,
- hedges are unavailable or expensive,
- order flow is one-sided,
- depth disappears,
- adverse selection rises,
- LULD/auction states make execution discontinuous.

5.7 Why RSP is not SPY

RSP is equal-weight S&P 500 exposure. SPY is cap-weight S&P 500 exposure. A hedge using SPY or futures is helpful but imperfect because factor and constituent weights differ.

$$\text{Basis Risk} = \Delta P_{RSP} - \beta \Delta P_{SPY}.$$

6 August 24, 2015 and RSP

6.1 High-level story

August 24, 2015 was a market-structure stress event. Broad risk-off pressure, futures disruption, delayed cash equity openings, Rule 48, LULD halts, retail market/stop orders, and market-maker quote withdrawal combined to cause ETFs to trade far away from estimated fair value.

SEC research notes severe volatility and extensive LULD halts on August 24, including in ETPs.³

6.2 Sequence

1. Global risk-off before U.S. open; index futures under pressure.
2. Futures hit limit/down conditions, hurting hedge confidence.
3. NYSE invoked historical Rule 48, relaxing certain pre-opening indication requirements during extreme volatility. Historical NYSE Rule 48 was designed to suspend some opening requirements in extreme market-wide volatility.⁴
4. Many stocks did not open cleanly at 9:30 because opening auctions faced large imbalances.
5. ETFs continued trading in some cases while many components were unopened, halted, wide, or stale.
6. Market makers could not confidently compute or hedge basket fair value.
7. They widened quotes and reduced displayed depth.
8. Retail stop-loss orders triggered and became market orders.
9. Market orders sold into thin ETF books, causing prints far below NAV/iNAV.
10. LULD pauses occurred in ETFs and stocks, creating asynchronous stop-start markets.
11. Prices re-anchored when constituents opened, hedges normalized, and liquidity returned.

³SEC Research Note, *Equity Market Volatility on August 24, 2015*: https://www.sec.gov/marketstructure/research/equity_market_volatility.pdf.

⁴NYSE Rule 48 interpretation: <https://www.nyse.com/publicdocs/nyse/markets/nyse/rule-interpretations/2007/07-110.pdf>.

6.3 RSP specifically

RSP is an equal-weight S&P 500 ETF. Its fair value depends on roughly equal-weighted constituent prices. During Aug 24, many constituents were not reliably open/tradable, so fair value was uncertain. RSP's ETF order book became thin, stop/market orders hit poor bids, and RSP printed far below estimated fair value before recovering.

Interview answer. RSP did not fall because the equal-weighted S&P 500 permanently lost 40% of fundamental value. It fell because the executable ETF market temporarily disconnected from estimated basket value. The fast recovery is the evidence: this was market structure and liquidity, not a normal fundamental repricing.

6.4 Why many stocks were not open at the beginning

Stocks require an opening process. The primary exchange must establish an opening price. Under huge sell imbalances, extreme volatility, and unclear reference prices, the opening auction can delay. If many components of an ETF are not open, the ETF's fair value must be estimated from stale closes, futures, ADRs, correlated ETFs, and partial prices.

6.5 Why market makers stepped back

A market maker quoting an ETF needs to know what it can hedge. If it buys ETF shares from panicked sellers, it must hedge by selling futures, SPY, or the basket. If those hedges are halted, wide, stale, or not tracking well, the market maker faces adverse selection and basis risk. So it widens spreads, reduces size, or only quotes defensively.

6.6 Why stop-loss orders were dangerous

A stop-loss order has a trigger price but, once triggered, it becomes a market order. If the book is thin, execution can occur far below the stop trigger. Stop-limit orders avoid terrible prices but can fail to fill.

6.7 Why trades were not necessarily cancelled

If orders executed according to exchange rules, a bad price does not automatically mean an erroneous trade. A market order accepts available liquidity. Unless a venue's clearly erroneous trade rules apply, participants can be stuck with the execution.

7 Anthony-style levered ETP stress scenario

7.1 Generic setup

Suppose Jane Street is lead market maker in a 3x long ETP on a liquid underlying. A client wants to sell a large block. Jane Street buys from the client, becoming long the ETP, and hedges by shorting the underlying basket, futures, or a related hedge.

Client sells 3x ETP

- > Jane Street buys ETP, long ETP exposure
- > Jane Street shorts basket/futures as hedge
- > underlier moves sharply
- > ETP may hit LULD band and pause
- > underlier may keep trading

7.2 Band calculation

For a Tier 2 3x ETP at reference price 30:

$$p = 10\% \times 3 = 30\%, \quad [21, 39].$$

If doubled window applies:

$$p = 10\% \times 3 \times 2 = 60\%, \quad [12, 48].$$

If the underlier is Tier 1 or Tier 2, calculate its own band separately. The ETP's LULD state is separate from the underlier's LULD state.

7.3 The key dislocation

The ETP and the underlier can be in different market states:

- ETP open, underlier halted/unopened: ETF fair value/hedge uncertain.
- ETP paused, underlier open: ETP cannot be traded, but hedge keeps moving.
- Both paused: less observable executable data.
- Hedge open but wide: tradable but expensive.
- Reopening auction indicative price moving: informative but not executable.

7.4 If the ETP is offered at lower band for 15 seconds

If NBO equals lower band and the limit state persists 15 seconds, the primary declares a trading pause. During the pause, continuous trading in the ETP stops. The underlier may continue trading.

7.5 Client still has 60% of block to sell: what do you know?

You know:

- Your filled ETP quantity.
- Your hedge fills and current hedge mark.
- The client's remaining block, if they tell you.
- Current underlier/futures/related ETF prices if those markets are open.
- Current LULD state, pause time, and expected auction process.
- Indicative reopening price/imbalance if disseminated.

You do not know:

- Final reopening auction price.
- Whether the auction will clear.
- How much liquidity will appear at reopening.
- Whether the client will keep selling aggressively.
- Whether other market makers/APs will re-enter.
- Whether the hedge remains optimal if the product is locked while the underlier moves.

Interview answer. During the pause, I know my filled ETP position and hedge position. I can observe and potentially trade the underlier, but I cannot trade the ETP. The real risk is basis and reopening risk: the ETF reopens at a price different from my hedge-implied fair value, while the client may still have flow to sell.

7.6 Desk exposure

For a long 3x ETP:

$$\text{ETP Economic Exposure} \approx 3 \times Q_{ETP} P_{ETP}.$$

If hedged with futures/basket:

$$\text{Net Exposure} = \text{ETP Economic Exposure} - \text{Hedge Exposure}.$$

If ETP is paused, the ETP side is not tradable but hedge side may be tradable. That creates:

- basis risk,
- reopening price risk,
- client flow risk,
- hedge over/under-adjustment risk,
- operational risk if orders/cancels are stuck.

7.7 Can you fill the client at indicative price?

Not as a normal market execution. Indicative price is not firm. You can negotiate a risk price, but then Jane Street is warehousing risk.

Interview answer. I would tell the client that indicative price is a useful estimate of reopening, but not an executable market price. If we provide liquidity during the halt, we need to price the reopening uncertainty, hedge uncertainty, and remaining block risk.

7.8 Why could the reopening auction fail?

- Indicative clearing price outside reopening collar.
- Sell imbalance too large.
- Insufficient buy liquidity.
- Market participants cancel/revise orders as underlier moves.
- Primary exchange system/auction issue.

If no reopening occurs within the packet's 10-minute path, prior reference price may remain the reference and bands can widen to 3x normal for 30 seconds before reverting.

7.9 Underlier down 8%, 3x ETP fair value down 24%

The first-order theoretical ETP return is:

$$R_{ETP} \approx 3r.$$

If $r = -8\%$, then $R_{ETP} \approx -24\%$ before fees, financing, compounding, and execution frictions.

But the interview question is not only the math. It is the market state. If the ETP cannot print below lower band while the underlier keeps moving, the executable ETP price can be pinned or paused while hedge value changes.

7.10 If the ETP is paused while underlier drops further

Jane Street long ETP and short underlier hedge. If underlier drops further:

- Short hedge makes money.
- Long ETP economic fair value should decline.
- But ETP cannot be traded, and reopening price is uncertain.
- If hedge not perfect, desk may be over- or under-hedged.

The risk is not just direction. It is whether the hedge-implied value and reopening price converge.

7.11 If underlier is halted while ETP still trades

Now ETF/ETP fair value becomes uncertain. A market maker may:

- widen ETP quotes,
- reduce displayed size,
- rely on futures/ADRs/similar ETFs,
- require larger edge for block trades,
- avoid market orders or aggressive quotes.

8 Leveraged and inverse ETP mechanics

8.1 Daily leverage target

A 3x long ETP targets approximately three times the daily return of the underlying index:

$$R_{ETP,daily} \approx 3R_{underlier,daily}.$$

It does not guarantee three times return over long horizons due to daily reset and compounding.

8.2 Volatility drag example

Underlier: +10% then -10%:

$$1.10 \times 0.90 = 0.99, \quad -1\%.$$

3x ETP: +30% then -30%:

$$1.30 \times 0.70 = 0.91, \quad -9\%.$$

Even if the underlier is nearly flat, the leveraged product loses more in a volatile path.

8.3 Daily rebalance pressure

For long leverage $L > 1$, starting NAV A , underlier return r :

- initial exposure = LA ,
- exposure after market move without rebalance = $LA(1 + r)$,
- target exposure after NAV changes = $LA(1 + Lr)$,
- rebalance amount = $LA[(1 + Lr) - (1 + r)] = LA r(L - 1)$.

For $L > 1$, the fund buys after up moves and sells after down moves to maintain daily leverage. This can matter near the close and in stressed markets.

8.4 Inverse products

Inverse products target negative daily returns, e.g. $L = -1$ or $L = -3$. Hedging and rebalance direction differs. The same framework applies: target exposure equals leverage factor times NAV.

9 New ETF product design prompt

Anthony mentioned candidates may be asked to design a new ETF product. This is not generic product fluff. You must design the product through market structure.

9.1 Framework

1. **Exposure:** what economic return does it target?
2. **User:** who wants the exposure and why?
3. **Construction:** physical basket, swaps, futures, options, mixed, or ETN?
4. **Liquidity:** are underlying components liquid enough?
5. **Hedgability:** can market makers hedge intraday?
6. **AP arbitrage:** creation/redemption in-kind or cash? Transparent basket?
7. **Market structure:** LULD, auctions, halts, settlement, borrow, FX, taxes.
8. **Risk:** tracking error, leverage, rebalance, concentration, volatility drag, model risk.
9. **Commercials:** AUM, spreads, fee, seed, market-maker participation.

9.2 Example: AI semiconductor 2x ETF

Exposure: 2x daily return to a basket of liquid semiconductor names.

Users: tactical investors who want concentrated AI hardware exposure without picking individual stocks.

Construction: swaps/futures where available plus physical holdings; daily reset; transparent methodology.

Market-structure questions:

- Are constituents liquid enough at open/close?
- Can APs create/redeem efficiently?
- Is there enough borrow/derivative capacity to hedge?
- What happens if top constituent halts?

- Does the product become too concentrated in one name?
- How wide should market makers quote under stress?
- What LULD band applies, especially with leverage?

Interview answer. I would not launch an ETF just because users want exposure. I would first ask whether the exposure is hedgeable and whether the arbitrage mechanism can keep the ETF tight in normal and stressed markets.

10 Risk concepts and calculations

10.1 Risk taxonomy

Risk	Meaning in these interviews
Market risk	Price moves against your position.
Liquidity risk	You cannot trade size without moving price materially.
Credit risk	Counterparty cannot perform.
Settlement risk	Cash/security transfer fails after execution.
Operational risk	Human/system/process error.
Model risk	Fair value, hedge, or risk model is wrong.
Basis risk	Hedge and hedged instrument diverge.
Regulatory risk	Trade/order violates rule or required process.

10.2 Position P&L

For linear position Q :

$$\Delta P\&L \approx Q\Delta P.$$

For notional $N = QP$ and return r :

$$\Delta P\&L \approx Nr.$$

For a 3x ETP, approximate economic exposure:

$$\text{Exposure} \approx 3 \times \text{ETP notional}.$$

10.3 VaR and expected shortfall

VaR at confidence α is the loss threshold exceeded with probability $1 - \alpha$:

$$VaR_\alpha = \inf\{x : \Pr(L \leq x) \geq \alpha\}.$$

Expected Shortfall is average loss conditional on being in the tail:

$$ES_\alpha = \mathbb{E}[L \mid L \geq VaR_\alpha].$$

Interview answer. VaR tells you a percentile loss. Expected Shortfall tells you how bad the tail is after that percentile is breached. In stressed trading questions, scenario tests are often more useful than VaR alone.

10.4 Stress scenarios to practice

- Underlier down 8% in 90 seconds.
- 3x ETP hits lower band and pauses.
- Top constituents unopened while ETF trades.
- Futures hedge limit-down pre-open.

- Client has 60% block remaining during halt.
- Reopening auction indicative price outside collar.
- ETF iNAV down 3% but market price down 35%.
- Spread widens from 2 bps to 100 bps.

10.5 Greeks

For options/nonlinear exposures:

Delta	sensitivity to underlying price.
Gamma	sensitivity of delta to underlying price.
Vega	sensitivity to implied volatility.
Theta	sensitivity to time passing.
Rho	sensitivity to interest rates.

10.6 Portfolio risk and concentration

For weights w and covariance matrix Σ :

$$\sigma_p^2 = w^\top \Sigma w.$$

A position can look small alone but large in portfolio terms if it is correlated with existing positions.

10.7 Risk display design, if asked conversationally

Even if not a full software design case, a trader risk display should prioritize:

- current position and P&L,
- large exposures by symbol/factor/counterparty,
- limit utilization,
- alerts/breaches,
- what-if shocks,
- hedge status,
- stale data warnings,
- drill-down to fills/orders causing the exposure.

11 Trade settlement risk

11.1 Definition

Execution means the trade is agreed. Settlement means securities and cash actually transfer. Settlement risk lives in the gap.

- Buyer must deliver funds.
- Seller must deliver securities.
- Risk: one side performs and the other does not.

The U.S. securities market moved to T+1 settlement for applicable securities transactions on May 28, 2024.⁵

11.2 Risk types inside settlement

⁵SEC Chair statement on T+1 implementation: <https://www.sec.gov/newsroom/press-releases/2024-62>.

Credit	Counterparty defaults or lacks funds/securities.
Liquidity	Counterparty cannot source cash/securities in time.
Market	Price moves while replacement trade is needed.
Operational	Wrong account, wrong ticker, mismatch, system error.
Legal/regulatory	Trade cannot be completed due to rule/eligibility issue.
Cross-border	FX/time-zone/local-market settlement mismatch.

11.3 Mitigations

- Delivery-versus-payment (DVP): cash and securities move together.
- Payment-versus-payment (PvP) for FX, e.g. Herstatt-style FX risk.
- Central clearing, netting, margin, default fund.
- Shorter settlement cycle.
- Reconciliation and affirmation.
- Fails management and buy-in/close-out processes.
- Counterparty limits, collateral, and exposure monitoring.

Interview answer. If asked to design from exchange/clearing perspective, focus on DVP, netting, margin, settlement matching, fails process, and default management. If asked from Jane Street perspective, focus on exposure limits, counterparty assessment, collateral, internal reconciliation, and replacement-risk hedging.

11.4 ETF creation/redemption settlement edge

For ETFs with international or less liquid components, ETF shares, underlying securities, FX, and derivatives may have different settlement calendars and time zones. APs manage this via cash components, collateral, cutoffs, FX, and operational buffers. The point is that creation/redemption is operationally real, not instantaneous abstract arbitrage.

12 Abnormal data and consistency puzzles

12.1 Core approach

Do not guess anomalies. Find invariants. An invariant is an accounting or lifecycle identity that must hold if data is correct.

$$\text{Final Cash} = \text{Initial Cash} + \text{Cash In} - \text{Cash Out}.$$

$$\text{Final Inventory} = \text{Initial Inventory} + \text{Buy Fills} - \text{Sell Fills}.$$

$$\text{Position}_t = \text{Position}_0 + \sum q_{\text{buy fills}} - \sum q_{\text{sell fills}}.$$

12.2 Order lifecycle invariants

- Filled quantity cannot exceed original quantity.
- Remaining = original - filled - cancelled.
- Actual position changes only on fills, not sends.
- Cancel confirm cannot happen before order exists.
- Fill after cancel confirm is suspicious unless sequence explains it.
- Duplicate execution ID should not be counted twice.
- Cash movement should equal price times quantity plus fees.
- Sell order must have inventory or permitted short/borrow/locate.

12.3 Example puzzle 1: inventory

Initial shares = 1,000. Trades: buy 200, sell 300, sell 100. Reported final inventory = 900.

Expected:

$$1000 + 200 - 300 - 100 = 800.$$

Reported 900 is wrong. Something is missing, duplicated, mis-signed, or one sell did not settle/fill.

12.4 Example puzzle 2: partial fill/cancel

Order: sell 1,000. Fills: 400 and 300. Cancel confirmed for remaining 400. This is impossible because:

$$1000 - 400 - 300 = 300, \quad \text{not } 400.$$

Either fill quantity, cancel quantity, or original order quantity is wrong.

12.5 Example puzzle 3: cash

Initial cash 10,000. Buy 100 shares at 50, fee 2. Sell 20 shares at 55, fee 1. Expected final cash:

$$10000 - 100 \times 50 - 2 + 20 \times 55 - 1 = 6097.$$

If reported final cash differs, check sign, fees, duplicate fill, missing fill, or wrong price.

12.6 Abnormal data system if they phrase it that way

If asked conceptually for months of inventory/orders/trades, start with deterministic checks before statistical anomaly detection:

1. Reconstruct positions and cash from event log.
2. Validate order lifecycle transitions.
3. Validate fills against original quantities.
4. Validate prices against NBBO/reference/fair value/LULD bands.
5. Validate timestamps and sequence order.
6. Then use distributional anomalies: unusual order rate, cancel ratio, slippage, stale data, position jumps.

Interview answer. The best first anomaly detector is not ML. It is a set of conservation laws: cash, inventory, order quantity, and event sequence. Statistical anomaly detection comes after the invariants.

13 SEC and market-structure rules

13.1 How to write or discuss an SEC rule

Use this structure:

1. Problem: what failure mode are we solving?
2. Scope: which securities, participants, and trading hours?
3. Definitions: key terms.
4. Requirements/prohibitions.
5. Exceptions.
6. Controls, supervision, records, and reporting.
7. Enforcement and penalties.
8. Phase-in and review.
9. Unintended consequences.

13.2 Reg NMS Rule 611 - Order Protection Rule

Rule 611 is the trade-through rule. Trading centers must establish, maintain, and enforce policies reasonably designed to prevent trade-throughs of protected quotations in NMS stocks, absent exceptions.⁶

Plain English: if a protected venue is displaying a better price, another venue generally should not execute at a worse price without a valid exception.

13.3 Reg SHO

Reg SHO governs short sales. Key concepts:

- marking sale orders long, short, or short exempt,
- locate requirement before many short sales,
- close-out requirements for fails to deliver,
- short-sale price test restrictions in certain decline scenarios,
- limited bona fide market-making exceptions.

SEC guidance summarizes Regulation SHO as having marking, price test, locate, and close-out requirements.⁷

13.4 Rule 15c3-5 - Market Access Rule

This is the anti-naked-access rule. Broker-dealers with market access must maintain risk management controls and supervisory procedures. The SEC describes the rule as addressing risks from automated rapid electronic trading strategies and effectively eliminating unfiltered or naked access.⁸

Core controls include:

- financial exposure limits,
- order size and price controls,
- erroneous and duplicative order prevention,
- regulatory compliance controls,
- regular review and direct control by broker-dealer.

13.5 LULD Plan

LULD prevents trades in NMS securities outside specified dynamic price bands and uses limit states/trading pauses to manage extraordinary individual-security volatility.

13.6 Rule 605 and Rule 606

- Rule 605: execution quality disclosures by market centers.
- Rule 606: order routing disclosures by broker-dealers.

Plain English: 605 helps evaluate execution quality; 606 helps understand routing practices and potential conflicts.

13.7 Historical NYSE Rule 48

Rule 48 was historically used to suspend certain pre-opening indication requirements during extreme market volatility. It matters in the Aug 24 story because fewer/later indications worsened uncertainty around opens. It is historical context, not necessarily a current live rule.

⁶17 CFR 242.611: <https://www.law.cornell.edu/cfr/text/17/242.611>; SEC Regulation NMS final rule: <https://www.sec.gov/files/rules/final/34-51808.pdf>.

⁷SEC Regulation SHO FAQ: <https://www.sec.gov/rules-regulations/staff-guidance/trading-markets-frequently-asked-questions-8>.

⁸SEC Rule 15c3-5 small entity compliance guide: <https://www.sec.gov/files/rules/final/2010/34-63241-secg.htm>.

14 High-yield scenario bank

14.1 Exchange scenarios

1. Build an exchange from scratch.
2. Explain price-time priority with two same-price orders.
3. Market order bigger than displayed depth.
4. Cancel arrives as fill arrives.
5. Order ACK lost; what should participant do?
6. Market data sequence gap: receive 100 then 102.
7. Opening auction has huge sell imbalance.
8. Reopening auction indicative price outside collar.
9. Closing auction attracts massive MOC imbalance.
10. Protected quote exists elsewhere; can venue execute at worse price?

14.2 LULD scenarios

1. Calculate Tier 1/Tier 2 bands above \$3.
2. Calculate 0.75–3 band.
3. Calculate leveraged ETP band.
4. Double band window at open/close.
5. Security stuck at lower band for 15 seconds.
6. Limit State cured before 15 seconds.
7. Straddle State vs Limit State.
8. Reopening auction fails.
9. No reopen within 10 minutes.
10. LULD pause in last 10 minutes before close.

14.3 ETF scenarios

1. ETF market price below iNAV: what should AP/MM do?
2. Underlying basket halted but ETF trades.
3. ETF halted but basket trades.
4. Futures hedge halted pre-open.
5. SPY open but RSP components not open.
6. ETF discount huge: why not free money?
7. ETF book thin after MMs pull quotes.
8. Stop-loss orders trigger into market orders.
9. iNAV stale due to unopened components.
10. Client wants block during halt at indicative price.

14.4 Risk and data scenarios

1. Cash in minus cash out does not match final cash.
2. Inventory equation fails.
3. Duplicate fill ID appears.
4. Fill quantity exceeds order quantity.
5. Cancel confirm before order ACK.
6. Price far from NBBO/fair value.
7. Portfolio risk increases even though position size is small.
8. VaR low but stress loss large.
9. Settlement fails after execution.
10. Counterparty cannot deliver security.

15 Rapid answer anchors

15.1 What happened to RSP?

Interview answer. RSP traded far below estimated fair value because the normal ETF arbitrage loop broke. Many underlying stocks were not open or were halted, futures hedges were stressed, LULD created asynchronous pauses, and market makers widened spreads/reduced depth. Stop and market orders sold into a thin ETF book, causing prints far below iNAV before the ETF re-anchored as constituents opened and hedges became usable.

15.2 Why did ETF and underlying prices diverge?

Interview answer. The ETF has its own order book and can trade even when parts of the basket are stale, wide, or halted. The underlying basket gives fair value, but only if the components are priceable and hedgeable. When that link weakens, the ETF order book can move independently, especially if market makers reduce liquidity.

15.3 How will ETF and underlying prices move relative to each other?

In normal markets, arbitrage pulls them together. If ETF is cheap, APs/MMs buy ETF and sell/short basket or hedge. If ETF is rich, they sell ETF and buy basket/create shares. In stress, convergence can be delayed because fair value is uncertain and hedge execution is risky.

15.4 Difference between ETF book and stock book

Interview answer. A stock book reflects direct supply/demand for one security. An ETF book is partly a synthetic book because liquidity providers quote it based on the underlying basket and hedges. So ETF displayed liquidity can be excellent in calm markets but disappear when the basket or hedge becomes uncertain.

15.5 What do you know during ETP LULD pause?

Interview answer. I know my fills, my hedge fills, current hedge markets if open, client remaining interest, LULD state, and indicative auction data if available. I do not know the final reopening price, whether reopening clears, or what flow appears. My exposure is filled ETP economic exposure minus hedge exposure, plus basis/reopening risk.

15.6 Why can reopening fail?

Interview answer. Because the auction's indicative clearing price can be outside collars, the imbalance can be too large, or there may be insufficient executable interest. The exchange may extend the pause and try again; under some rules, if no reopen occurs within a specified time, other trading centers may resume with widened bands.

15.7 What is settlement risk?

Interview answer. Settlement risk is the risk that after a trade is executed, one side delivers cash or securities but the other side fails to perform. The mitigations are DVP/PvP, central clearing, margin/collateral, shorter settlement cycles, reconciliation, and exposure limits.

15.8 How to find abnormal data?

Interview answer. Start from invariants: final cash equals initial cash plus inflows minus outflows; final inventory equals initial inventory plus buys minus sells; order remaining equals original minus filled minus cancelled. Then identify which event is missing, duplicated, mis-signed, or misordered.

16 Last 48-hour checklist

1. Recite exchange architecture without notes.
2. Work through auction clearing example.
3. Calculate LULD bands for Tier 1, Tier 2, low-price, and 3x ETP.
4. Explain Limit State, Straddle State, 15-second cure, five-minute pause.
5. Explain reopening auction and why indicative price is not executable.
6. Explain AP creation/redemption and why it can fail.
7. Explain RSP August 24 in three minutes.
8. Explain client block + ETP pause: know/don't know/exposure.
9. Practice 20 abnormal data invariant puzzles.
10. Practice SEC rule explanations in plain English.

17 Deep concept clarifiers: do not mix these up

17.1 Price objects that sound similar but are not

Object	Meaning and interview use
Last sale	The most recent reported trade. It is factual but can be stale or unrepresentative after a dislocated print.
Best bid/offer	Current displayed executable quotes at the top of book. They show immediate displayed liquidity, not fundamental value.
NBBO	Best protected displayed bid/offer across venues. Important for Reg NMS and routing.
Midpoint	Average of bid and offer; useful quick reference, but can be meaningless if spread is huge.
Reference price	Rule input for LULD bands. It is not necessarily fair value.
Indicative auction price	Estimated auction clearing price if auction ran now. Useful but not executable until auction prints.
iNAV/IIV	Intraday estimate of ETF basket value. Can be stale if components are stale/unopened.
NAV	Official fund value, usually end-of-day.
Fair value	A trader/MM model estimate of what the instrument should be worth given tradable hedges and component prices.
Executable price	Price at which actual liquidity can trade now. This is what matters for P&L.

Trap. In stress, the most common error is to treat a non-executable estimate as if it were an executable market price. iNAV, indicative price, fair value, and last sale can all differ.

17.2 Price movement vs value movement vs liquidity movement

When something trades down, ask three separate questions:

1. Did fundamental/economic value change?
2. Did executable market price move because of liquidity/order-book pressure?
3. Did the observed quote/trade become unreliable because of stale data or halted components?

For RSP on Aug 24, economic basket value did not fall 40%. The ETF executable price temporarily fell far more because liquidity vanished and market orders hit weak bids.

18 Detailed exchange scenarios and model answers

18.1 Scenario: market order walks the book

Book:

Offers

100 @ 50.10

200 @ 50.20

700 @ 50.50

A market buy for 500 executes:

- 100 @ 50.10
- 200 @ 50.20
- 200 @ 50.50

Average price:

$$\frac{100(50.10) + 200(50.20) + 200(50.50)}{500} = 50.30.$$

Interview answer. The order did not execute at the midpoint or fair value. It consumed the actual offers in the book. This is why thin depth matters more than a headline last price.

18.2 Scenario: limit order crossing existing book

Book has best offer 50.10 for 100 shares. New buy limit arrives: buy 200 at 50.20. It first trades 100 at 50.10. If next offer is 50.20 for at least 100, it trades 100 at 50.20. If no more offers at or below 50.20, remaining quantity rests as buy 100 at 50.20.

18.3 Scenario: two orders at same price

If two sell orders are both at 50.00, earlier exchange sequence has priority. If a buy order arrives for only enough quantity to fill one of them, earlier order fills first. This is why queue position has value.

18.4 Scenario: modify and queue priority

If a resting buy order at 50.00 changes to 50.10, it usually loses priority because it is economically a new price. If it only reduces size, some venues may preserve priority because it reduces liquidity rather than improving it. Safest interview answer: exact rule is venue-specific; model price changes as cancel/re-enter unless told otherwise.

18.5 Scenario: hidden liquidity and displayed depth

Displayed depth may understate total available liquidity because of hidden/iceberg/dark liquidity, but in a crisis it may overstate practical liquidity because quotes can vanish or be cancelled. Never assume displayed depth is a guarantee of future liquidity.

18.6 Scenario: smart order routing and trade-through

If Exchange A shows 100 shares offered at 50.00 and Exchange B shows 100 shares offered at 50.05, a broker/router generally should not route a normal buy order to execute at 50.05 while ignoring the protected 50.00 offer, absent an exception. That is the purpose of the Order Protection Rule.

19 Auction math: clearing price examples

19.1 Simple opening auction

Suppose auction orders are:

Buy interest	Sell interest
Market buy 300	Market sell 200
Buy 500 @ 100	Sell 200 @ 99
Buy 200 @ 99	Sell 500 @ 100
Buy 300 @ 98	Sell 400 @ 101

At candidate price 99:

- executable buys: market buy 300 + buy 500 @100 + buy 200 @99 = 1000
- executable sells: market sell 200 + sell 200 @99 = 400
- executable volume = 400

At candidate price 100:

- executable buys: market buy 300 + buy 500 @100 = 800
- executable sells: market sell 200 + sell 200 @99 + sell 500 @100 = 900
- executable volume = 800

So 100 is better than 99 under a max-volume rule. Tie-breakers vary by venue.

Interview answer. An auction is not just the midpoint. It is a batch-clearing mechanism. It chooses a clearing price from the supply/demand in the auction book according to published rules.

19.2 Opening auction delay

If the indicative open is far away from previous close, if imbalance is huge, or if there is not enough contra-side interest, the exchange may delay. For an ETF, delayed constituent opens create stale inputs for basket fair value.

19.3 Reopening auction under LULD

After a LULD pause, participants submit/cancel/revise orders. The exchange disseminates indicative price and imbalance. If the indicative price is inside collars and there is enough interest, it reopens. If not, the pause can extend.

19.4 Closing auction and ETF/index pressure

The close is important because index funds and ETFs benchmark to closing prices. Leveraged ETPs also rebalance to target daily leverage, often creating end-of-day demand to buy or sell the underlying/derivatives after large moves.

20 LULD edge-case drill

20.1 Reference-price update example

Current reference price is 100. The pro-forma five-minute average is 100.80. Move = 0.80%, so no update if the threshold is 1%.

If pro-forma is 101.20, move = 1.20%, so reference updates to 101.20. New bands are calculated from 101.20. A new reference price usually remains in effect for a minimum time before updating again.

20.2 Limit state cured before 15 seconds

If NBO equals lower band, the market enters Limit State. If the limit quotation is executed or cancelled before 15 seconds and the market moves away from the band, trading continues and no pause occurs.

20.3 Limit state persists 15 seconds

If the market remains at the band, the primary declares a pause. The security stops continuous trading and later reopens through auction.

20.4 Straddle state scenario

If lower band is 95 and NBB is 94 while NBO is 96, the market is not necessarily in a normal executable state. It may be in Straddle State, and the primary can pause if trading is not normal.

20.5 LULD and market orders

LULD does not guarantee the investor avoids a bad execution. A market order can execute at poor prices within the band before a pause. LULD prevents prints outside bands, not all prints far from fair value.

20.6 LULD and ETF fair value

ETF lower band could be calculated from ETF reference price while basket fair value is somewhere else. If ETF trades down toward lower band but basket fair value is stable, LULD only sees the ETF's own price mechanics. It does not know that the ETF is cheap versus the basket.

20.7 LULD and asynchronous markets

The hardest cases:

- ETF open, constituents halted.
- ETF paused, constituents open.
- ETF reopened, some constituents still unopened.
- Futures paused but ETF/stocks open.
- Components reopen at different times.

In all cases, arbitrage is impaired because the ETF and hedges are not simultaneously executable.

21 ETF arbitrage and convergence scenarios

21.1 ETF cheap to basket

Suppose fair value is 100 and ETF trades at 98.

Normal arbitrage:

1. Buy ETF at 98.
2. Sell/short basket, futures, or proxy hedge around 100 exposure.
3. If possible, redeem ETF shares for basket or unwind when convergence occurs.

Expected pressure:

- Buying ETF pushes ETF price up.
- Selling basket/hedge pushes underlying/proxy down.
- Discount narrows from both sides, though usually more through ETF price if underlying is deeper.

21.2 ETF rich to basket

Suppose fair value is 100 and ETF trades at 102.

Normal arbitrage:

1. Sell/short ETF at 102.
2. Buy basket or hedge exposure.
3. Create ETF shares if needed to cover short / deliver.

Expected pressure: ETF price down, basket/proxy up, premium narrows.

21.3 Why the order books differ when convergence happens

When ETF is cheap, ETF book may show one-sided selling and weak bids because market makers demand edge. The stock/basket books may remain deeper because natural buyers/sellers and index liquidity are larger. The arbitrage trade connects the two books, but the link is not frictionless. ETF book liquidity is often conditional on basket liquidity.

21.4 Premium/discount is not enough information

A 2% discount could be attractive or dangerous. Need to know:

- Is fair value reliable?
- Are components open and liquid?
- Can hedge be executed?
- How much size can be traded?
- Are creation/redemption windows open?
- What are borrow, funding, and settlement constraints?

22 August 24: answers to likely follow-ups

22.1 Was it a flash crash?

Yes, it was flash-crash-like in the sense of a rapid, temporary, severe price dislocation and recovery. But be precise: it was not only one stock crashing; it was a broad market open stress where many ETFs and stocks experienced dislocations, halts, and liquidity breakdowns.

22.2 Did LULD help or hurt?

Both. It helped by preventing unlimited trades through empty books and forcing pauses. It hurt or complicated the event by creating asynchronous halts/reopenings across ETFs and constituents, so the arbitrage loop was less reliable.

22.3 Why did RSP recover?

Because the underlying constituents did not permanently lose the amount implied by the ETF low prints. As components opened, hedges became usable, and market makers regained confidence, ETF quotes returned closer to basket fair value.

22.4 What should a retail investor have done differently?

Use limit orders, avoid market orders at stressed open, understand stop-loss orders can become market orders, and check ETF price versus iNAV/fair value/related products before panicking.

22.5 What should a market maker do differently?

Monitor component open/halt status, fair-value confidence, LULD proximity, hedge availability, and auction states. Quote smaller/wider when confidence is low. Avoid providing large guaranteed liquidity unless compensated for risk.

22.6 What system could reduce harm?

A dislocation warning system that flags ETF price versus fair value, component halt ratio, spread/depth, LULD state, and hedge availability. For users, default to limit orders or require confirmation for market orders in high-risk states.

23 More detailed levered ETP numerical cases

23.1 Case: client block and hedge

Assume:

- 3x long ETP reference price = 30.
- Client wants to sell 1,000,000 shares.
- Jane Street buys 400,000 shares before pause.
- Client has 600,000 shares remaining.
- ETP notional bought = 12,000,000.
- Economic underlier exposure $\approx 3 \times 12,000,000 = 36,000,000$ long.
- Desk shorts 36,000,000 of underlier/futures as hedge.
If underlier falls 8% and ETP fair value falls 24%:
- Hedge P&L $\approx +36,000,000 \times 8\% = +2,880,000$.
- ETP fair-value loss $\approx -12,000,000 \times 24\% = -2,880,000$.
- Perfect hedge roughly offsets.

But if ETP is paused, the offset is model-based. The actual reopening price may differ from hedge-implied fair value. The client still has 600,000 shares to sell, which can push reopening flow.

23.2 Case: underlier down 12%, ETP theoretical down 36%

Reference price 30. Tier 2 normal 3x band is [21,39]. Theoretical fair value after -12% underlier move is:

$$30(1 - 0.36) = 19.20.$$

This is below the lower band 21. The ETP cannot simply print continuously at 19.20 while band is 21. It may sit at lower band and trigger pause. Underlier can keep trading, so hedge/fair-value relationship keeps moving while ETF is locked.

23.3 What if hedge is imperfect?

If the desk hedges RSP with SPY or futures, exposure is not one-to-one. Suppose RSP beta to hedge is 0.95, but equal-weight factors underperform cap-weight factors during stress. The hedge may fail exactly when needed.

$$\text{Residual P\&L} = \Delta P_{ETP} - \beta \Delta P_{hedge}.$$

23.4 What if client insists on immediacy?

If ETF is halted, the desk cannot execute an exchange trade in the ETF. It can negotiate principal risk transfer, but that is not the same as an exchange execution. The price must include:

- indicative auction uncertainty,
- hedge basis risk,
- remaining block pressure,
- reopening failure risk,
- market-impact cost,
- balance sheet/inventory cost.

24 Regulation mini-answers and rule-writing templates

24.1 If asked: why is Rule 611 an SEC rule?

Because without trade-through protection, a venue could execute an order at a worse price even when a better protected quote is displayed elsewhere. The rule tries to make displayed quotes meaningful across fragmented venues and gives participants confidence that protected best prices cannot simply be ignored.

24.2 If asked: write a market access rule

Problem: unfiltered access allows fat-finger, duplicate, excessive, or non-compliant orders to hit markets.

Rule skeleton:

1. Applies to broker-dealers with direct or sponsored market access.
2. Broker-dealer must maintain pre-trade risk controls under its direct control.
3. Controls must reject orders exceeding size, price, notional, credit, or capital thresholds.
4. Controls must prevent duplicative or clearly erroneous orders.
5. Controls must ensure compliance with regulatory requirements before order entry.
6. Broker-dealer must regularly test, document, and supervise controls.

Tradeoff: too strict blocks legitimate liquidity; too loose creates market-integrity risk.

24.3 If asked: write a better ETF dislocation rule

Potential design:

1. Require ETF-specific warnings when ETF price deviates from iNAV/fair value by more than threshold and component confidence is low.
2. Do not automatically halt just because ETF diverges; divergence may be fair if basket is stale.
3. Require brokers to warn or default to limit orders for retail market orders in high-risk ETF states.
4. Publish component halt ratio and fair-value confidence.
5. Review unintended consequences: warnings may reduce liquidity or create herding.

24.4 If asked: write an LULD amendment

Use the rule-writing frame:

- Objective: reduce ETF-specific dislocations without stopping valid price discovery.
- Scope: NMS ETPs during regular trading, especially opens/reopens.
- Inputs: ETF bands, component halt ratio, basket confidence, hedge-market status.
- Requirement: enhanced auction/retail warnings when component confidence below threshold.
- Exception: highly liquid broad ETFs with robust hedge availability.
- Metrics: bad executions, false warnings, reopening success, spreads, liquidity withdrawal.

25 Mini mock interview: 60-second answers

25.1 Build an exchange

An exchange receives orders, validates them, sequences them, matches crossing orders in an order book, and publishes market data and private execution reports. I would build gateway, validation, sequencer, matching engine, order book, event log, market-data publisher, and execution-report publisher. The key invariant is deterministic price-time priority and correct order lifecycle state.

25.2 Walk through an opening auction

Before continuous trading begins, orders accumulate. The exchange calculates an indicative clearing price using buy/sell interest. It generally chooses a price that maximizes executable volume, with tie-breakers like imbalance and reference price. If imbalance is too large or price uncertainty too high, opening can be delayed.

25.3 What happened on Aug 24?

Broad market stress hit the open. Futures were stressed, many stocks opened late, NYSE invoked Rule 48, LULD halts hit many instruments, and ETF market makers could not confidently price/hedge baskets. They widened spreads and reduced depth. Retail market and stop orders then hit thin ETF books, causing ETF prints far below fair value before recovery.

25.4 Why can ETF arbitrage break?

Because arbitrage requires reliable fair value and executable hedges. If components are halted/unopened, futures are paused, spreads are wide, or creation/redemption is operationally delayed, the apparent discount is not risk-free.

25.5 How do you calculate LULD band for levered ETP?

Find tier and price bucket, take base percentage, multiply by leverage factor, then apply time-of-day multiplier if in widened window. Example Tier 2 3x ETP at 30 normal time: 10% times 3 = 30%, so [21,39].

25.6 What happens if reopening auction fails?

The pause extends and the primary attempts to reopen again. In the packet's 10-minute failure path, if no reopen occurs within 10 minutes or there is a systems issue, other venues may resume trading with prior reference price and temporarily widened bands.

25.7 Why can indicative price not be used as execution price?

Because it is an estimate of where the auction would clear if run now. It is not a firm bid or offer. It can move as orders change, and final auction price may differ.

25.8 What is the difference between market price and fair value?

Market price is where actual trades occur. Fair value is an economic estimate based on underlying inputs. During stress, market price can move far from fair value because liquidity and hedging capacity break down.

25.9 How would you detect bad data?

Start with invariants: cash equation, inventory equation, order lifecycle equation, and sequence validity. Then identify whether a fill is missing, duplicated, mis-signed, mispriced, or misordered.

26 Long-form mock prompts to practice out loud

1. You are asked to rebuild Nasdaq from scratch. Explain actors, order types, order book, matching engine, auctions, market data, and edge cases.
2. RSP trades down 40% while iNAV is down 3%. Explain every mechanism that can create this.
3. A 3x ETP is halted at the lower band. The client has 60% of a block remaining. What do you know, what do you not know, and what is your exposure?
4. An ETF is cheap versus basket, but 30% of basket weight is unopened. Would you arbitrage? What changes your answer?
5. Design a new ETF product around a current theme. Explain not just demand but construction, hedging, creation/redemption, LULD, settlement, and launch metrics.
6. Given trade/order/inventory data with a mismatch, find the error using invariants.
7. Explain Reg NMS Rule 611, Reg SHO, Market Access Rule, LULD, and Rule 605/606 in plain English.
8. Explain settlement risk from buyer, seller, exchange/clearing, and Jane Street perspectives.

27 Common wrong answers and better answers

Weak answer	Better answer
ETF price was wrong.	ETF executable price was real but temporarily disconnected from basket fair value.
LULD caused the crash.	LULD interacted with volatility and asynchronous market states; it was not the root cause.
APs would arbitrage it instantly.	Arbitrage requires reliable fair value, hedge, funding, and operational capacity.
Indicative price is where we can trade.	Indicative price estimates auction clearing; it is not executable until the auction prints.
A cancel means order is gone.	Cancel is pending until exchange confirms; fill can arrive first.
Market order gets fair price.	Market order gets available liquidity, which may be far from fair value.

RSP should track S&P 500 exactly.

Just look at NAV.

Risk equals VaR.

Settlement risk is impossible on exchange.

RSP is equal-weighted; SPY/futures are imperfect hedges.

NAV is end-of-day; intraday fair value/iNAV can be stale or uncertain.

VaR is one metric; stress, ES, liquidity, concentration, and basis risk matter more in these scenarios.

Exchange/clearing reduces it, but settlement timing, fails, operational breaks, and counterparty exposure still matter.
